



DEALER BEST PRACTICE SERIES

Major Component
Installation Kits Improve
R&I Process

Application Maintenance Site Management **Component Rebuild** Safety MARC Management

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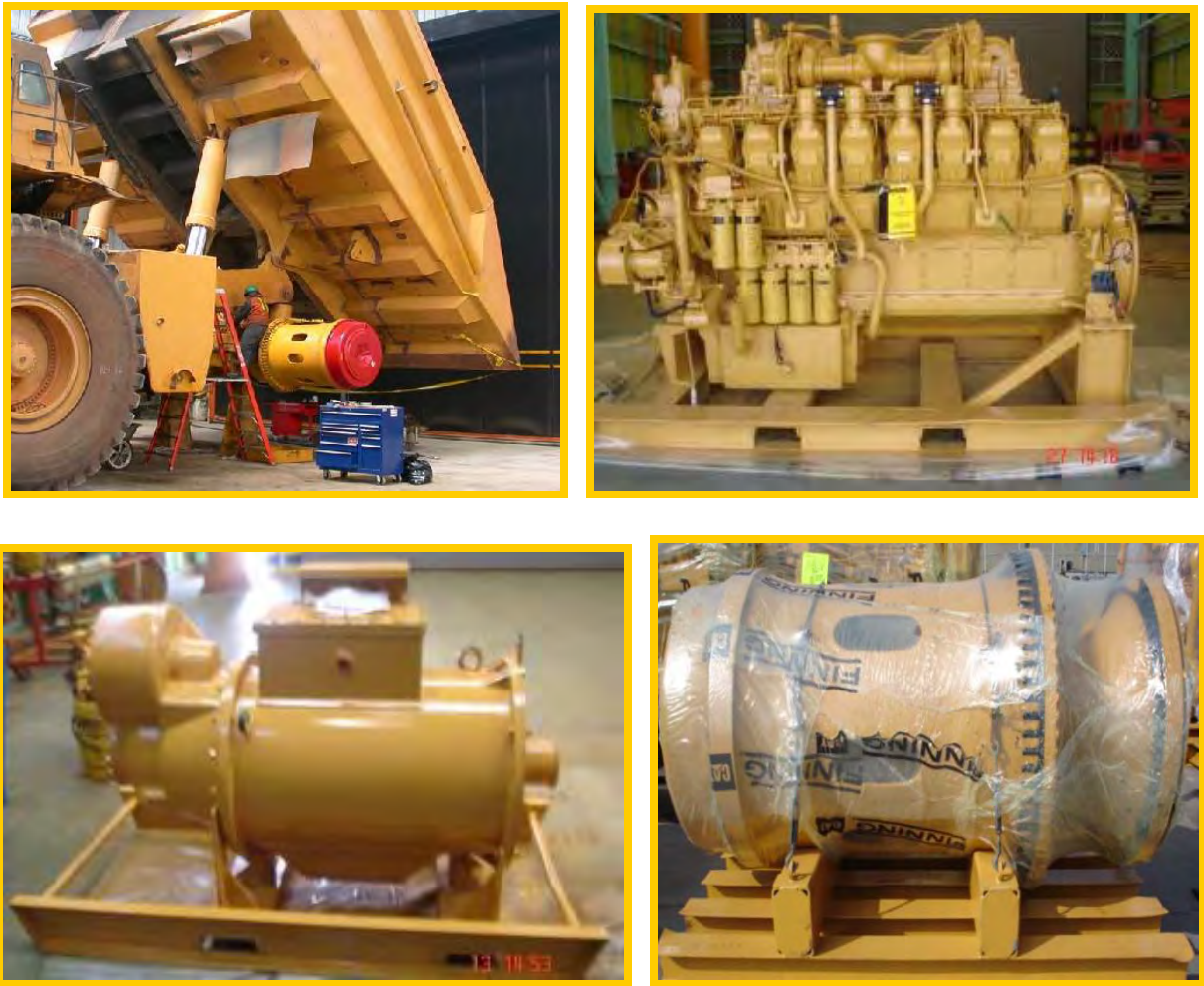
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1.0 Introduction

Proper installation of major components into a machine can have substantial impact on component life and reliability. A high quality installation procedure requires a well-defined process utilizing the right tools, parts, and readily available technical information.

This Best Practices discusses the use of standardized component installation “kits” that include all parts, control checklists, and other technical information required to perform a major component installation.

Standardized install kits help insure that component installations are performed in a consistent, efficient, high quality manner to deliver optimum component life.



Various Major Component Install Kits

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2.0 Best Practice Description

Many dealers, working with their customers, have jointly developed a standardized component installation process along with a required parts Bill of Materials (RBOM) for the installation. The required Bill of Materials is often delivered in the form of a “kit”, which can either be shipped along with the component from the dealer shop or assembled on the mine site.

The important benefit of this Best Practice is that everything needed to perform a job professionally and efficiently is staged with the component when a major component is installed in a machine.



Engine installation standard parts kit and steering hoses

3.0 Implementation Steps

There are typically two parts of a component installation kit. The first is a Required Bill of Materials kit containing all the parts required to perform the component installation. The second is a literature package containing checklists, technical information, and other required forms and documents.

Bill of Materials

Every component installation requires some parts. Caterpillar R & I checklists list many of these required parts. Typically, parts include standard as well as break-in High Efficiency fluid filters, mounts and mount hardware, air intake system seals, coolant system hoses and clamps, drive shaft bolts, or any other required hardware.

In addition, many dealers customize the installation parts kits to meet specific customer needs as related to machine application or customer preference. Many customers, for example, want to change all compartment fluid hoses whenever a component is removed as standard practice.

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The best way to manage installation Bill of Materials is to develop a spreadsheet organized by machine model and arrangement. Each component would then have a specific list of parts for the installation. In this way, the parts list is created one time and used every time a component matching the model an arrangement is rebuilt. Once the list is developed, the dealer should ensure part numbers remain current.

Shown below is a sample Bill of Materials used by one dealer. Note that this example is several years old and part numbers may not be fully current.

COMPONENT	PART LIST	SERIAL	MODEL	PART N.	DESCRIPTION	QTY
TRANSMISION	K03-793CTR	4AR	793C	1H8128	SEAL-O-RING	1
TRANSMISION	K03-793CTR	4AR	793C	1R0719	FILTER A	1
TRANSMISION	K03-793CTR	4AR	793C	1R0741	ELEMENT AS-H	2
TRANSMISION	K03-793CTR	4AR	793C	2H6338	SEAL	2
TRANSMISION	K03-793CTR	4AR	793C	2J7761	RING	1
TRANSMISION	K03-793CTR	4AR	793C	2K4472	O RING	1
TRANSMISION	K03-793CTR	4AR	793C	2N2486	BOLT	8
TRANSMISION	K03-793CTR	4AR	793C	2S8439	SEAL O RING	1
TRANSMISION	K03-793CTR	4AR	793C	3D2824	SEAL O RING	1
TRANSMISION	K03-793CTR	4AR	793C	3J1907	SEAL	1
TRANSMISION	K03-793CTR	4AR	793C	4J0522	SEAL O-RING	4
TRANSMISION	K03-793CTR	4AR	793C	4J0527	SEAL O RING	2
TRANSMISION	K03-793CTR	4AR	793C	5D6428	SEAL O RING	1
TRANSMISION	K03-793CTR	4AR	793C	5K9090	SEAL O RING	6
TRANSMISION	K03-793CTR	4AR	793C	5P8182	GROMMET	3
TRANSMISION	K03-793CTR	4AR	793C	5P8444	GROMMET	1
TRANSMISION	K03-793CTR	4AR	793C	6D9157	SEAL	1
TRANSMISION	K03-793CTR	4AR	793C	6G0078	BREATHER	1
TRANSMISION	K03-793CTR	4AR	793C	6J2680	SEAL O RING	2
TRANSMISION	K03-793CTR	4AR	793C	6V8398	SEAL O RING	4
TRANSMISION	K03-793CTR	4AR	793C	8R7036	SPIDER G	2
TRANSMISION	K03-793CTR	4AR	793C	1259460	HOSE AS	1
TRANSMISION	K03-793CTR	4AR	793C	1259461	HOSE AS	1
TRANSMISION	K03-793CTR	4AR	793C	1259463	HOSE AS	1
TRANSMISION	K03-793CTR	4AR	793C	1280928	HOSE AS	1
TRANSMISION	K03-793CTR	4AR	793C	1308433	GROMMET	2
TRANSMISION	K03-793CTR	4AR	793C	1315525	HOSE AS	1
TRANSMISION	K03-793CTR	4AR	793C	1380580	HOSE AS	1
TRANSMISION	K03-793CTR	4AR	793C	1419450	HOSE AS	1
TRANSMISION	K03-793CTR	4AR	793C	1419451	HOSE AS	1

Attached to this Best Practice is a sample spreadsheet of the installation kits used by one dealer. This information is intended as an example and part numbers many not be current.

Information Package

A typical information package for a component installation includes:

- Caterpillar R & I Checklist (See Attachments for index and checklists)
- R & I Special Instructions
- Oil Sample Kit
- Forms required for the installation record keeping and component tracking.

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Removal & Installation Checklist - Off Highway Trucks

Caterpillar Form No.	01-040514-00 (PC)	Models 777, 785, 789, 793	- Differential
Caterpillar Form No.	01-040515-00 (PC)	Models 777, 785, 789, 793	- Engine & Torque Converter
Caterpillar Form No.	01-040516-00 (PC)	Models 777, 785, 789, 793	- Final Drive
Caterpillar Form No.	01-040517-00 (PC)	Models 777, 785, 789, 793	- Torque Converter
Caterpillar Form No.	01-040518-00 (PC)	Models 777, 785, 789, 793	- Transmission
Caterpillar Form No.	01-040519-00 (PC)	Model 797	- Differential
Caterpillar Form No.	01-040520-00 (PC)	Model 797	- Engine & Torque Converter
Caterpillar Form No.	01-040521-01 (PC)	Model 797	- Final Drive
Caterpillar Form No.	01-040522-00 (PC)	Model 797	- Torque Converter
Caterpillar Form No.	01-040523-00 (PC)	Model 797	- Transmission

In addition, some dealers have developed special plastic cards with special instructions to attach to components.



4.0 Benefits

- Creates a standardized R & I process
- Increases R & I efficiency – less machine downtime
- Ensures correct R & I process
- Improves component reliability
- Improves component life
- Develops customer relations

5.0 Resources Required

- Time to develop Bill of Materials and process
- Materials for assembling Component Install Kits

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6.0 Supporting Attachments / References

Attached to this PDF:

[Component Install Kits.xls](#)

[R&I Checklists in SIS \(PDF files\)](#)

Sample Bill of Materials lists

A collection of R&I checklists in PDF format

Note: The file "Index and Form Numbers.xls" contains a listing of the applicable models for each checklist.

Special Instructions

REHS0715 Removal and Installation of Engine and Torque Converter 785-793

REHS0716 Removal and Installation of Transmission 785 – 793

REHS0717 Removal and Installation of Differential and Final Drive 785-793

Guidelines

SEBF1017 Improving Component Durability – Component Removal and Installation

7.0 Related Best Practices

None.

8.0 Acknowledgements

This Best Practice was written by:

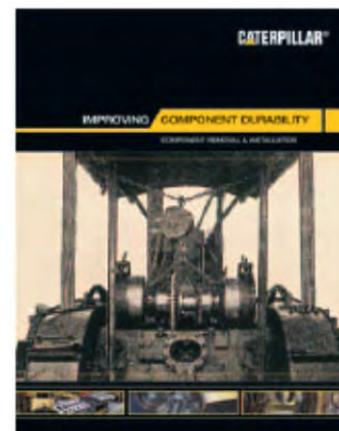
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